

GROUP ASSIGNMENT (2026.02)

COURSE CODE: UECM3993
COURSE: AM, AS, FM

Course Title: PREDICTIVE MODELLING
DEPARTMENT: DMAS

Instructions

1. This is a group assignment with **four** to **seven** students including a **group leader** per group.
2. The **group leader** need to submit the following items through email (liewhh@utar.edu.my) or MS Teams Chat:
 - a list of members (with signatures)
 - group title/name (cannot be too bizarre or offensive)
 - the dataset of interest from the given listfor documentation before the start of assignment (Week 4).
3. The lecturer reserves the right to assign remainder students who are not part of any assignment group to an assignment group with less than **seven** members. Those who cannot form a group before Week 4 may be penalised.
4. Towards the deadline, the **group leader** is responsible to submit the following documents for the group assignment through email (liewhh@utar.edu.my) or MS Teams Chat:
 - (a) “Group Name” report.pdf Wednesday of Week 11
 - (b) “Group Name” program code(s) Wednesday of Week 11
5. **Deadline of submission** for **group assignment report** and **group programming code** is 6pm, 15 April 2026 (Wednesday of Week 11).
6. **Group Presentation** will be scheduled in weeks 11 to 13, date and time to be announced. Each presentation is limited to a maximum of 22 minutes (5 groups per 2-hour lecture).
7. In the case of **late submission** for the report and program script, 10% of the maximum marks will be deducted if the work is up to one day late (24 hours) and additional 10% of the maximum marks for each of the subsequent days.
8. **Plagiarism or purely AI generated contents are not allowed.** If the works are found to be plagiarised or 100% AI generated report, no marks will be awarded and the incident will be reported to the university for further action (you may be suspended if you are found guilty).
9. The group assignment report **is recommended to** contain the information on the **contributions of members to the project** in ratio or percentage.

Marks

- Marks will be equally distributed by default. If the group assignment report has a section on **individual contributions** (either in the first page or second page or the appendix), each member will receive

$$\text{teamwork marks} \times \left(1 - 0.4 \times \frac{\max \text{IC} - \text{IC}}{\max \text{IC}}\right)$$

where **IC** = individual contribution. For example, (Note: the same contribution can be applied to programming code.)

- A group with 7 members with contributions (20%, 20%, 20%, 20%, 4%, 3%, 3%) and the report is 13 out of 18
 - * 4 members will get $13 \times \left(1 - 0.4 \times \frac{20-20}{20}\right) = 13$ marks
 - * 2 members will get $13 \times \left(1 - 0.4 \times \frac{20-3}{20}\right) = 8.58$ marks
 - * 1 member will get $13 \times \left(1 - 0.4 \times \frac{20-4}{20}\right) = 8.84$ marks
- A group with 4 members with contributions (A:10%, B:20%, C:30%, D:40%) and the report is 15 out of 18:
 - * member A gets $15 \times \left(1 - 0.4 \times \frac{40-40}{40}\right) = 15$ marks
 - * member B gets $15 \times \left(1 - 0.4 \times \frac{40-30}{40}\right) = 13.5$ marks
 - * member C gets $15 \times \left(1 - 0.4 \times \frac{40-20}{40}\right) = 12$ marks
 - * member D gets $15 \times \left(1 - 0.4 \times \frac{40-10}{40}\right) = 10.5$ marks

The rationale for the mark adjustment is to promote active participation in group assignment and to prevent individual members from doing nothing in the group. Any member who does absolutely nothing will only receive 60% of the teamwork marks.

- Each member will usually receive equal marks for the group programming code indicating active learning in the programming tasks unless the **group leader** reports that some members contributes nothing to the group programming. In such cases, the above rules will be used to downgrade the marks of the members who did not contribute to the group programming.
- Each member will receive equal marks for the group oral presentation with extra marks for members who present really well unless the **group leader** wants to have a different weights for the group members.
- A group leader can be **re-elected** if more than half of the members are not happy with the group leader at least one week before the submission of the assignment.

Group Assignment Report (18%)

1. Pick a dataset from the following list and perform a **case study** on the dataset by applying the **unsupervised learning** and **supervised learning** on the dataset:
 - Assessing Mathematics Learning in Higher Education Dataset (<https://archive.ics.uci.edu/dataset/1031/dataset+for+assessing+mathematics+learning+in+higher+education>): In this case study, you will analyse an education data with 7 features and 1 output for mathematics learning which is difficult to fit because the features, despite categorical are too large. You need to perform additional investigation on the education systems in European countries to understand why certain features stand out rather than performing superficial predictive analysis and evaluation.
 - Gallstone Dataset (<https://archive.ics.uci.edu/dataset/1150/gallstone-1>): The dataset is in Excel format and being zip twice. In this case study, you need to implement a program which can work with the original data in Excel format to learn the significant features which contributes to gallstone. The more we learn how to predict different illnesses can allow us to build a better healthcare system that provides early detection of illnesses and hopefully supports early prevention.
 - e-Commerce with the Online Shoppers Purchasing Intention Dataset (<https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset>): In this case study, you will try to learn the features that are significant to purchase intention relevant to e-Commerce. The data is time-based and is rather large (> 10k rows) and you will need to choose a proper sampling strategy perform predictive analysis and evaluation.
 - Turkish Crowdfunding Startups Dataset (<https://archive.ics.uci.edu/dataset/1025/turkish+crowdfunding+startups>): The dataset is collected by Turkey in 2022. It includes various characteristics such as crowdfunding projects, project descriptions, targeted and raised funds, campaign duration, and number of backers. In this case study, you will analyse the data to learn the important features. You will need to translate the Turkish words (e.g. `basari_durumu`) into English (e.g `success_status`) before you perform predictive analysis. You may use Google translate or AI tools to help you but make sure you document and cross check your results. The purpose of this case study is to learn from Turkish experience to help Malaysia grow its entrepreneurs using crowdfunding.
 - Dry Bean Dataset (<https://archive.ics.uci.edu/dataset/602/dry+bean+dataset>): Models should be developed to identify 7 different registered classes of dry beans in this case study. In this case study, you should think critically by investigating how the tabular data is obtained from the original data and how modern AI may perform better with the original data.
 - Audit Dataset (<https://archive.ics.uci.edu/dataset/475/audit+data>): Exhaustive one year non-confidential data in the year 2015 to 2016 of firms is collected from the Auditor Office of India to build a predictor for classifying suspicious firms. In this case study, the data is already split into the training data `audit_risk.csv` and the testing data `trial.csv`. However, the two data have different features. So it is a challenging problem in which you need to clean the two data (with duplicate

rows and inconsistent columns) to make sure they are consistent. The purpose of this case study is to learn that real world data requires cleaning and finding suspicious firms is a challenging problem.

2. 0.5 out of 18 marks are allocated for the registration of data for a case study. 0.5 will be awarded to the first and second groups that register the same title. The third group to register the same title will receive 0.3, the fourth group and onwards will receive 0 to prevent too many groups working on the same case study. Self-proposed data may be allowed upon lecturer's approval but the group will also receive 0 out of 0.5 marks. Those who change data after week 4 will also receive 0 out of 0.5 marks. Those who did not form and/or join an assignment group and are being assigned by the lecturer will also receive 0 out of 0.5 mark. This is to promote active learning, i.e. students need to take initiative to form a group and work on a case study.
3. By using appropriate statistical software framework (R, Python with Scikit-learn, WEKA, C++, etc.), build models with the different statistical learning approaches (both unsupervised and supervised learning methods) which are covered in this course (and those which are not covered, but descriptions and documentations are required for methods not introduced in lecture with good references), find the "best" model for the data set selected to meet proper research objectives, i.e. the research objectives should not be superficial task like 'trying out predictive models' but should be linking to the application of data in real-world problem. Your program script must work on the **original data** from the URL or a small portion of the marks (0.5–1 out of 18 marks) will be deducted.
4. The report should be written in appropriate report format which is simple, neat and easy to refer and containing the following contents:
 - An introduction with appropriate references and proper research objectives.
 - The understanding of the data is presented nicely with appropriate academic citations.
 - Unsupervised learning with EDA of the features and various advanced methods to identify interesting patterns from the data.
 - Supervised learning with various predictive models related to the data. The performance measures should be summarised appropriately for comparison.
 - A conclusion.

Group Programming Code (10%)

1. Write a programming code (or nicely structured programming codes) with an appropriate use of libraries which analyse the **original dataset** from the URL which is picked in the group assignment report and works in a data science pipeline. Marks are deducted if Excel or other expensive software (e.g. SAS, SPSS) is used to pre-process the data rather than automating the data processing in the programming code.
2. Marks will be **deducted** if the programming code is in notebook and/or markdown and/or non-text formats which are not directly executable.
3. Marks may be **deducted** if data processing taught in the practical are not used but the sophisticated techniques from the Internet are copied (such as dplyr, etc.) without proper documentation in the assignment report or the programming code.
4. The programming code can only use free and legal software. The default is R (and Python). The group who try other free and legal open source software (such as Java, C++) which are cross-platform and does not have too much dependencies, i.e. the program can run on Microsoft Windows (of various versions), GNU/Linux platform, MacOS/X, etc., will receive extra marks.
5. The programming code(s) need to demonstrate the appropriate use of **supervised** and **unsupervised** learning with free and legal statistical software tool and appropriate comments.

Group Oral Presentation (10%)

1. Prepare presentation slides which summarises the group assignment report and possible future improvements.
2. An oral presentation which involves every member or a presentation by just one or few representative member(s) are allowed.
3. The oral presentation should cover the following aspects:
 - A good description of the problem and a systematic use of unsupervised and supervised learning methods to discover important information from the dataset.
 - A good illustration of unsupervised learning and supervised learning results.
 - **Explain the algorithm** behind the best model with respect to following aspects:
 - The mathematical/statistical idea behind best supervised learning model for the problem.
 - Explain how the model would be updated if new data comes in.
 - The presentation is well-timed (less than 22 minutes but not too short, heavy marks may be deducted for over-time presentation), has a proper conclusion and is interesting (not reading literally from the slides).